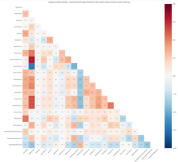
Customer Churn Prediction & Retention Analysis

1. Executive Summary

Core Objective
Define target (Churn) and remove leakage variables. Drop identifiers and redundant features (e.g., Total Charges, CLTV). CLTV validated via correlation with existing variables.

Preprocessing:
- Numerical: median imputation + standardization
- Categorical: mode imputation + one-hot encoding

2. Data Processing



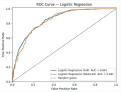
(a) Correlation Heatmap

(b) VIF

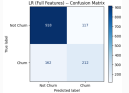
Low–moderate correlations indicate weak direct relationships. VIF reveals multicollinearity from one-hot encoding and overlapping features.

3. Model Performance

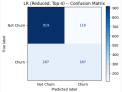
3.1 Logistic Regression



(a) ROC Curve



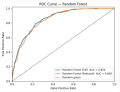
(b) Full Features



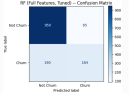
(c) 4 Features

The ROC curves show that the full model achieves a slightly higher AUC (0.847 vs 0.841), indicating better discrimination ability. The confusion matrices also suggest that the full model identifies more churn cases, while the reduced model misses more positives.

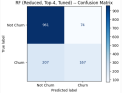
3.2 Random Forest



(a) ROC Curve



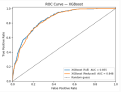
(b) Full Features



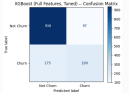
(c) 4 Features

Full model: Acc 0.805, Rec 0.492, F1 0.572, AUC 0.854
Reduced model: Acc 0.801, Rec 0.447, F1 0.543, AUC 0.849
Full model detects more churn (184 vs 167), with fewer misses.

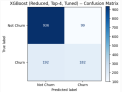
3.3 XGBoost



(a) ROC Curve



(b) Full Features



(c) 4 Features

Using the optimized parameters, we train two models on different feature sets. With the full feature set, the model achieves
ROC-AUC of 0.855. With the reduced feature set, performance

4. Final Conclusion

Model	Accuracy	Precision	Recall	F1
Logistic Regression (Full)	0.8020	0.6444	**0.5988**	**0.6031**
Logistic Regression (Reduced)	0.7850	0.6172	0.5000	0.5524
Random Forest (Full)	0.8048	**0.6840**	0.4920	0.5723
Random Forest (Reduced)	0.8006	0.6929	0.4465	0.5431
XGBoost (Full)	**0.8070**	0.6723	0.5321	0.5940
XGBoost (Reduced)	0.7935	0.6477	0.4866	0.5557

Among all six experiments, XGBoost with the full feat
achieves the highest ROC-AUC (0.8549), indicating th
discriminative ability.

Compared with Logistic Regression and Random Fores
consistently delivers stronger performance, especially i
non-linear relationships and improving recall for churn

Additionally, the full feature set is preferred, as it pres
available information and allows the model to leverage
interactions among variables, which is critical for accu
predicting customer churn.

Therefore, XGBoost (Full Features) is selected as the
as it provides the best balance of predictive performan
comprehensive feature utilization.